



Security by Orchestration: Slice Deployment with Latency and Isolation Constraints

Yannick Carlinet (speaker), Laurent Valeyre, Nancy Perrot, Jean-Philippe Wary

Orange Innovation

November 29, 2023



Dedicat-6G

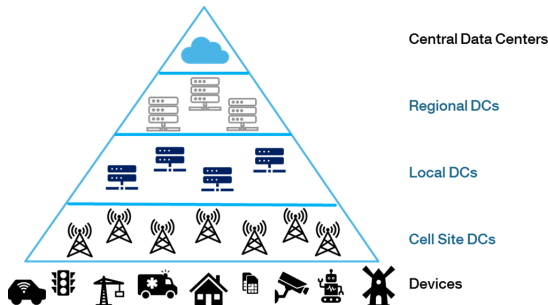


 Project Coordinator	 Technical Manager		
			
			
			

- European collaborative project
- Objectives: develop a platform
 - Beyond 5G
 - Coverage extension
 - Distribution of intelligence between devices, edge, and cloud
 - Support for B5G/6G services (ultra-low latency, broadband, ...)
- Infos and deliverables: <https://dedicat6g.eu/>

Slicing

- Virtual networks that are allocated resources depending of the requirements of the services, e.g.,
 - Mobile Broadband
 - Massive IoT (Internet of Things)
 - Ultra-Reliable Low-Latency Communication (URLLC)



Edge computing

- Service deployment in edge network
 - Reduced latency
 - Reduction of data transmitted to the cloud
- Challenges
 - Resource constraints
 - Security
 - A compromised VM can compromise other VMs in the same physical host
 - e.g. exploits can allow a VM to access forbidden RAM addresses
 - data stealing, data poisoning, ...
 - Physical isolation of slices much too costly
 - Orchestration
 - Management of resources across several edge nodes and clusters
- We propose a multi-cluster orchestrator
 - Enables deployment of service chains
 - Provides security isolation

Problem Statement

- Given a network topology (complete graph)
- Deployment of VNFs (Virtual Network Function) on nodes
- Migration of already placed VNFs
- VNFs compose Service Chains
- Service Requirements, e.g.,
 - Resources (e.g., cpu, memory, storage)
 - Maximum acceptable communication latency
 - Security isolation
- Costs
 - VNF Migration
 - Node usage (energy, cloud provider contract, ...)
- **How to deploy/migrate VNFs at minimum cost ?**

Resource requirements

Variables	
x_{uf}	Binary indicator of function f placed on node u
y_u	Binary indicator of node u in use

Sets	
V	Set of nodes (index u or v)
F	Set of functions (index f or g)
Constants	
R_f^{cpu}	Requirement in vCPU of function f
C_u^{cpu}	Capacity in vCPU of node u
R_f^{ram}	Requirement in RAM of function f
C_u^{ram}	Capacity in RAM of node u
R_f^{stor}	Requirement in storage of function f
C_u^{stor}	Capacity in storage of node u

$$\sum_{f \in F} R_f^{cpu} x_{uf} \leq C_u^{cpu} y_u \quad \forall u \in V$$

$$\sum_{f \in F} R_f^{ram} x_{uf} \leq C_u^{ram} y_u \quad \forall u \in V$$

$$\sum_{f \in F} R_f^{stor} x_{uf} \leq C_u^{stor} y_u \quad \forall u \in V$$

Latency Model

- Communication latency
- Induced by links used by VNFs of the chain

Variables	
x_{uf}	Binary indicator of function f placed on node u
y_u	Binary indicator of node u in use

Sets	
V	Set of nodes (index u or v)
F	Set of functions (index f or g)
S	Set of chains (index k)
F_k	Set of functions in chain $k \in S$
Constants	
l_{uv}	Latency induced by using link (u, v)
L_k	Maximum end-to-end latency admitted by chain k
S_{fg}	Indicator whether f and g are consecutive

$$\sum_{u,v \in V} \sum_{f,g \in F_k} x_{uf} x_{vg} S_{fg} l_{uv} \leq L_k \quad \forall k \in S$$

Security Isolation Model

- Part of SLA between IaaS provider and Service provider
- Each VNF has an Isolation level
 - Requirement of the service (i.e. chosen by Service provider)
 - E.g. high level if sensitive data processing
- ... and a Trust level
 - Defined by IaaS provider (e.g. security audit)
 - Represents the difficulty to compromise the VNF
- Note: Isolation level cannot be set above Security level
- Two VNFs can be on the same host iff. $T_1 \geq I_2$ and $T_2 \geq I_1$

Security Isolation Model

Variables	
x_{uf}	Binary indicator of function f placed on node u
y_u	Binary indicator of node u in use

Sets	
V	Set of nodes (index u or v)
F	Set of functions (index f or g)
Constants	
I_f	Isolation level of function f
T_f	Trust level of function f

$$I_f(x_{uf} + x_{uf'} - 1) \leq T_{f'}x_{uf'}$$

$$\forall u \in V, \forall f, f' \in F$$

Cost Model

- Cost for selecting a location (migration)
- Cost for using a node

Variables	
x_{uf}	Binary indicator of function f placed on node u
y_u	Binary indicator of node u in use

Sets	
V	Set of nodes (index u or v)
F	Set of functions (index f or g)
Constants	
c_u	Cost of using node u
c_{uf}	Cost of migrating/placing f to node u

$$\sum_{u \in V} c_u y_u + \sum_{u \in V} \sum_{f \in F} c_{uf} x_{uf}$$

Formulation

Sets

V	Set of nodes (index u or v)
F	Set of functions (index f or g)
S	Set of chains (index k)
F_k	Set of functions in chain $k \in S$

Constants

I_f	Isolation level of function f
T_f	Trust level of function f
R_f^{cpu}	Requirement in vCPU of function f
C_u^{cpu}	Capacity in vCPU of node u
R_f^{ram}	Requirement in RAM of function f
C_u^{ram}	Capacity in RAM of node u
R_f^{stor}	Requirement in storage of function f
C_u^{stor}	Capacity in storage of node u
c_u	Cost of using node u
c_{uf}	Cost of migrating/placing f to node u
l_{uv}	Latency induced by using link (u, v)
L_k	Maximum end-to-end latency admitted by chain k
S_{fg}	Indicator whether f and g are consecutive
B_{fg}	Indicator whether g is a backup for f

Variables

x_{uf}	Binary indicator of function f placed on node u
y_u	Binary indicator of node u in use

$$\min \sum_{u \in V} c_u y_u + \sum_{u \in V} \sum_{f \in F} c_{uf} x_{uf}$$

subject to

$$\sum_{f \in F} R_f^{cpu} x_{uf} \leq C_u^{cpu} y_u \quad \forall u \in V$$

$$\sum_{f \in F} R_f^{ram} x_{uf} \leq C_u^{ram} y_u \quad \forall u \in V$$

$$\sum_{f \in F} R_f^{stor} x_{uf} \leq C_u^{stor} y_u \quad \forall u \in V$$

$$I_f(x_{uf} + x_{uf'} - 1) \leq T_{f'} x_{uf'} \quad \forall u \in V, \forall f, f' \in F$$

$$\sum_{u \in V} x_{uf} = 1 \quad \forall f \in F$$

$$\sum_{u, v \in V} \sum_{f, g \in F_k} x_{uf} x_{vg} S_{fg} l_{uv} \leq L_k \quad \forall k \in S$$

$$x_{uf} + x_{ug} \leq 2 - B_{fg} \quad \forall u \in V, \forall f, g \in F$$

Latency constraint linearization

- New set F^C : set of couples of consecutive functions
- New variable: $z_{uvfg} \in \{0, 1\} \quad \forall u, v \in V, \forall (f, g) \in F^C$
- Latency constraints become:

$$\sum_{u, v \in V} \sum_{(f, g) \in F^C} z_{uvfg} l_{uv} \leq L_k \quad \forall k \in S$$

$$x_{uf} + x_{vg} - 1 \leq z_{uvfg} \quad \forall u, v \in V, \forall (f, g) \in F^C$$

$$z_{uvfg} \leq x_{uf} \quad \forall u, v \in V, \forall (f, g) \in F^C$$

$$z_{uvfg} \leq x_{vg} \quad \forall u, v \in V, \forall (f, g) \in F^C$$

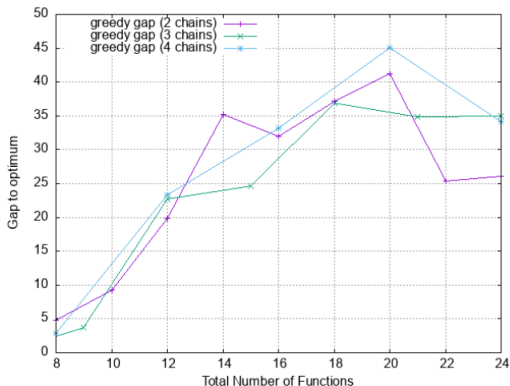
- z_{uvfg} takes the value 1 if $x_{uf} = x_{vg} = 1$, and the value 0 otherwise

Multi-criteria optimization

- SLSCP (Security and Latency Sensitive Service Chaining Problem)
- Objective function: min cost
- Alternate objective: min overall latency
- Two-steps process
 - Minimize latency, then minimize cost with constrained latency
 - Minimize cost, then minimize latency with constrained cost

Experimental Results

- Implementation in Python / Pyomo
 - ILP Formulation
 - Greedy heuristics (chains sorted by decreasing latency, or random)



Conclusion

- Main contributions
 - SLSCP problem description and modeling
 - Algorithm enables multi-provider cloud orchestration
 - Set-up of a demo with Kubernetes and AWS clusters (Amazon Web Services)
- Future work
 - Improve performance wrt. execution time
 - Integration of GCP clusters in demo (Google Cloud Computing)

Thank you
Questions ?



DEDICAT 6G

